Gioia Zheng

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Profile

Undergraduate at Sapienza University of Rome. Building MS MARCO retrieval-augmented QA end-to-end on a single 6-core CPU machine — BM25, dense (SBERT/FAISS), cross-encoder rerank, T5 generation — with paired-bootstrap CIs, NLI-entailment grounding, and a regression failure taxonomy. Interested in the gap between surface-form and semantic-faithfulness metrics in RAG evaluation.

Education

Sapienza University of Rome — Sep 2022 – Present
B.Sc. in Applied Computer Science and Artificial Intelligence

Experience

Web Development Intern — Mar 2026 – Present
Idearia — Rome, Italy (On-site)

- Developed and maintained PHP / WordPress backend systems for client-facing websites.
- Built n8n automation workflows integrating WordPress forms and content with external services, reducing manual operations.

NLP Algorithm Engineer Intern — Feb 2026 – May 2026
Microsoft — Natural Language Computing (Remote)

- Headline result on MS MARCO `dev/small` (6,980 queries, 8.8M-passage corpus): replacing BM25 first-stage with dense top-100 reranked by a cross-encoder, same frozen T5-small generator, lifted Token-F1 **0.197** → **0.368** and ROUGE-L **0.193** → **0.368**. 95% paired-bootstrap CIs on per-query Δ (10k resamples, seed 42) strictly above 0 across all four surface-form metrics.
- Pipeline I built and own end-to-end: `bm25s` sparse (MRR@10 0.170, in line with Anserini's 0.184; gap consistent with tokenizer differences), dense `all-MiniLM-L6-v2` over a FAISS `IndexFlatIP` on L2-normalised embeddings, `cross-encoder/ms-marco-MiniLM-L-6-v2` over the dense top-100 (538k pairs at ~32 pairs/s, 4h37m wall-clock on 6 cores), T5-small generation.
- Ruled out a surface-form artefact with a DistilBERT-BERTScore proxy on 3,000 paired qids: semantic-proxy Δ +0.173 sits within 0.002 of the surface-form Token-F1 Δ, well inside encoder noise. Chose DistilBERT over canonical `roberta-large` as an explicit CPU-cost vs. signal tradeoff, documented in the report.
- 40-query deterministic-rule triage of the 233-query regression bucket: **~90% of regressions are generator-side truncation** (mid-word or $\leq$ 3-token outputs), not retrieval failure. Falsified the decode-budget reading by re-running full `dev/small` at `max_new_tokens=64`→`128`: truncation share moved 90% → 87.5%, all four rerank Δs moved < 0.005. The output style is intrinsic to T5-small on this prompt format, not a budget symptom.
- NLI-entailment grounding (`cross-encoder/nli-deberta-v3-small`, 3,000-pair subsample) is the only metric whose paired Δ reverses sign: -0.145 (95% CI $[-0.16, -0.13]$, $p < 0.001$). Hypothesis: fragmentary mid-word outputs that inflate token overlap cannot be entailed by a sentence-level NLI cross-encoder. Flagged as a generator-swap follow-up rather than a regression to roll back.
- Reproducibility scaffold: per-run manifest (git SHA, config hash, dependency-file hashes, environment fingerprint), pinned `requirements-lock.txt`, ~295-test `pytest` suite, CI on every push, batch-eval design with no hidden one-shot CLI.

Projects

Handwritten OCR pipeline — github.com/GioiaZheng/handwritten-ocr-system
PyTorch — residual CNN + BiLSTM + CTC, IAM dataset

- CRNN for variable-length handwritten line transcription: 9 residual conv blocks (32→128 channels, batchnorm + leaky-ReLU + dropout 0.2), 2 stacked BiLSTM layers (hidden 256 / 64), linear head over vocab + CTC blank.
- Training and held-out evaluation with per-batch CER and WER (`torchmetrics`) streamed to TensorBoard; greedy CTC decode at inference; preprocessing pipeline (grayscale, normalisation, augmentation).

LeetCode solutions and pattern notes — github.com/GioiaZheng/Leetcode-Solutions
Python — schema-validated repo, GitHub Actions CI

- 98 problems, one directory each, with per-problem metadata (id, difficulty, topics, status) validated against a JSON schema. Topic index and catalog regenerated on every push; CI fails if the generated docs drift out of sync with the source metadata.
- Quality gate on every push: `ruff` lint, `pytest` on a curated subset, structural validation, matrix CI on Python 3.11 and 3.12.

Chat service — github.com/GioiaZheng/go-chat-system
Go backend + Vue frontend + SQLite

- Go REST backend with SQLite for single-writer durability; Vue frontend embedded into the Go binary at build time so the service ships as one static artifact.
- OpenAPI schema checked in CI; per-push matrix: `gofmt`, `go vet`, unit tests, OpenAPI lint, frontend build.

Skills

Primary	Python, PyTorch, `bm25s`, FAISS, sentence-transformers, cross-encoder reranking, paired-bootstrap CIs, BERTScore / NLI-entailment grounding, pytest

Working	Go, Docker, Linux, GitHub Actions CI, OpenAPI, SQLite, LaTeX
Familiar	JavaScript / React Native, Vue
Languages	English (Professional), Italian (Bilingual), Chinese (Native), German (Beginner)